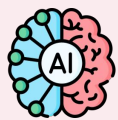


PhysicsMind

Problem & Goal

Input Video + Query



Foundational Model



VQA Understanding



World Model

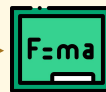


Video Generation

Dataset Construction



Physics TextBook



Concept Formulation



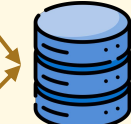
Variable Selection



Real-World Recording



Simulator



Raw Data

Dataset Annotation & Quality Control



Raw Data



Questions Designed by Experts



Answered by Experts

Alignment
Clarity
Physics
Validity
Accuracy

Diverse Scenarios

Center of Mass

Lever Equilibrium

Newton's First Law

Real + Sim

Dynamic & Controlled Conditions

Varying Object Shapes

Varying Masses

Varying Positions

Varying Setup

PhysicsMind

Directional Consistency

Final State Accuracy

Seg. Mask IoU + Center diff

velocity accel. similarity

Physics-Aware Metrics

Spatial Alignment

Unified model Evaluation

Fine-grained Temporal Metrics

Quantitative Reasoning